

1. Administrative & Study Identification

Full Study Title: A Phase 3, randomized, active-controlled, double-blind clinical study to evaluate the efficacy and safety of MK-8527 oral once-monthly as HIV-1 preexposure prophylaxis
Short Title/Acronym: EXPrESSIVE-11 (MK-8527-011) **Protocol Number:** MK-8527-011 **NCT Number:** NCT07044297 **Sponsor Name:** Merck Sharp & Dohme LLC (MSD) **Investigational Product:** MK-8527 (oral nucleoside reverse transcriptase translocation inhibitor [NRTTI]) **Phase of Development:** Phase III **Medical Monitor:** MSD Clinical Operations Lead / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of oral MK-8527 taken once a month to prevent HIV-1 infection, compared to the standard daily active-comparator (tenofovir disoproxil fumarate/emtricitabine [F/TDF]). **Secondary Objectives:** To evaluate the safety, tolerability, and pharmacokinetics of MK-8527, specifically regarding its ability to maintain sustained protective drug levels over a one-month dosing interval.

2.2 Background & Rationale To address the urgent global need for additional, user-friendly HIV pre-exposure prophylaxis (PrEP) options. With only an estimated 18% of global PrEP need currently met, taking a daily pill poses adherence, access, and stigma challenges for many. A once-monthly oral PrEP regimen like MK-8527 may provide a discreet, convenient, and effective alternative to improve global HIV prevention implementation.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (daily tenofovir disoproxil fumarate/emtricitabine [Truvada or generic equivalents]) **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 4,400 participants **Number of Sites:** Global (approx. 16 countries)
Estimated Study Start: 31-Jul-2025 **Estimated Primary Completion:** 22-Jul-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 16 years (enrolling sexually active adults and adolescents). 2. People without HIV who have a higher likelihood of HIV exposure (including gay men, transgender women and men, and nonbinary people). 3. Ability to understand the purpose of the study and provide signed informed consent.

4.2 Exclusion Criteria 1. Known current or prior infection with Human Immunodeficiency Virus Type 1 (HIV-1). 2. Prior use of cabotegravir, lenacapavir, or any other long-acting HIV PrEP product in the past. 3. Clinically significant medical conditions or laboratory abnormalities that would compromise participant safety or data interpretation.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: 11 mg MK-8527, dosed once-monthly (every 4 weeks). **Route of Administration:** Oral. **Duration of Treatment:** Up to 2 years.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant medications, such as oral hormonal contraceptives (e.g., levonorgestrel/ethinyl estradiol), which have been demonstrated to have no significant drug-drug interaction with MK-8527. **Prohibited:** Use of concurrent or prior long-acting antiretroviral therapies for PrEP (e.g., cabotegravir, lenacapavir).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, HIV-1 Testing, Safety Labs
Baseline	Day 0	Randomization, First Dose of MK-8527 or Active Comparator
Interim	Every 4 Weeks	Study Drug Dispensation, HIV-1 Testing, Safety Assessments, Adherence Review
End of Study	Up to 2 Years	Final Efficacy/Safety Assessment, Exit Interview, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: HIV-1 infection incidence rate (efficacy of once-monthly MK-8527 compared against daily oral F/TDF) over the treatment period. **Measurement Method:** Confirmed HIV-1 positive diagnosis via Central Laboratory Analysis.

7.2 Secondary & Exploratory Endpoints * Safety and tolerability profile (incidence of Adverse Events and Serious Adverse Events, including monitoring of CD4 T cell and total lymphocyte counts).

* Pharmacokinetic parameters (e.g., trough concentrations, C_{max}) to confirm sustained protective drug levels over a 28-day dosing interval.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via patient interviews and safety labs, with a specific focus on CD4/lymphocyte counts, headache, nausea, and fatigue. **Serious Adverse Events (SAEs):** Reported to the sponsor within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee (IDMC) will review safety and efficacy data at pre-specified intervals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) Set for primary efficacy (HIV incidence) and Safety Set for AEs/SAEs. **Sample Size Justification:** Target sample size of $\approx 4,400$ participants powered adequately ($\alpha = 0.05$) to establish non-inferiority or superiority of MK-8527 against the active comparator (daily F/TDF) in a high-incidence population. **Handling of Missing Data:** Incidence rates will be analyzed accounting for patient-years of follow-up; missing HIV-1 status data will be handled via predefined censoring or imputation strategies as per the SAP.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to ensure protocol compliance and data integrity. **Audit Trail:** Data entry and query resolution will be maintained via a validated electronic Data Capture (EDC) system with full audit trails.

11. Study Identifiers & Governance

Primary ID: MK-8527-011 **Registry Number:** NCT07044297 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC (MSD) **Study Title:** EXPrESSIVE-11 **Official Regulatory Title:** A Phase 3, randomized, active-controlled, double-blind clinical study to evaluate the efficacy and safety of MK-8527 oral once-monthly as HIV-1 preexposure prophylaxis

12. Clinical Framework (HIV PrEP Specific)

Primary Condition: HIV-1 Pre-exposure Prophylaxis (PrEP)

Diagnosis Threshold: Confirmed HIV-1 negative status at baseline with documented high likelihood of HIV exposure. **Medical**

Methodology: Administration of NRTTI (MK-8527). **Optimization**

Target: Maintenance of therapeutic drug concentrations to block HIV reverse transcriptase translocation for ≥ 28 days.

13. Study Procedures & Methodology

Management Pathway: Routine clinical monitoring for HIV PrEP

adherence. **Education Protocol (HCP):** Site initiation and training on proper dispensing of study intervention and blinding

procedures. **Education Protocol (Patient):** Safe sex practices, medication adherence counseling, and regular HIV testing. **Quality**

Audit Mechanism: Tracking of patient adherence to the monthly dosing schedule and daily comparator pill intake.

14. Observation & Follow-Up Schedule

Total Duration: Up to 104 weeks (2 years) **Onsite Visit Cadence:**

Every 4 weeks for monthly dosing **Remote Monitoring Frequency:**

Intermittent per site standard practice. **Checklist Review**

Interval: Regular review of adherence diaries and AE occurrences at every monthly visit.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				